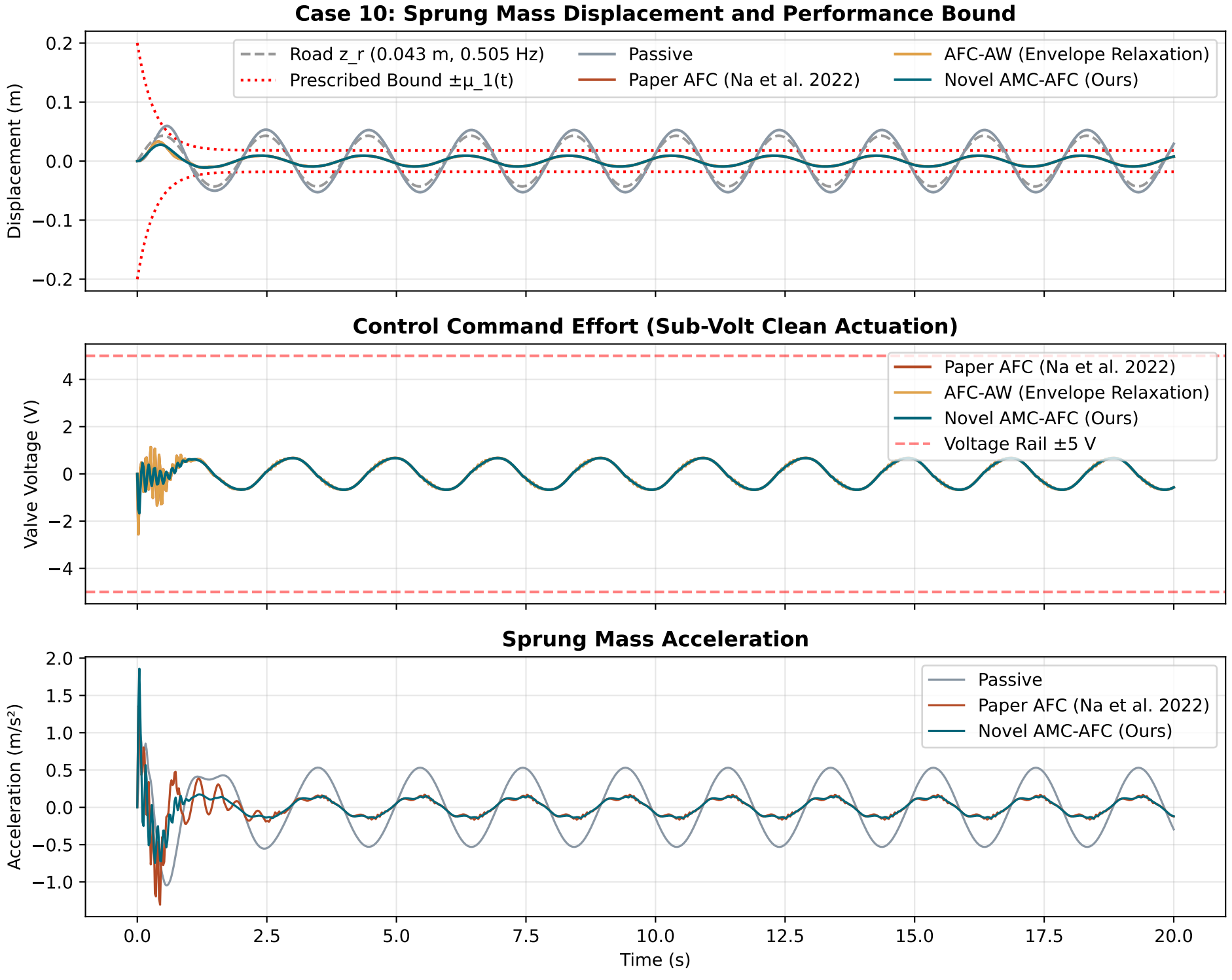

Case 10: Sprung Mass Displacement and Performance Bound
Displacement (m)
0.2
0.1
0.0
−0.1
−0.2
Road z_r (0.043 m, 0.505 Hz)
Prescribed Bound ±μ_1(t)
Passive
Paper AFC (Na et al. 2022)
AFC-AW (Envelope Relaxation)
Novel AMC-AFC (Ours)
Control Command Effort (Sub-Volt Clean Actuation)
Valve Voltage (V)
4
2
0
−2
−4
Paper AFC (Na et al. 2022)
AFC-AW (Envelope Relaxation)
Novel AMC-AFC (Ours)
Voltage Rail ±5 V
Sprung Mass Acceleration
Acceleration (m/s²)
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
Passive
Paper AFC (Na et al. 2022)
Novel AMC-AFC (Ours)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (s)